Susana Loureiro

Software Engineer | ML

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Summary

Software Engineer with a strong foundation in full-stack development, systems design, and machine learning, currently completing a BSc in Informatics Engineering at ISEP. Experienced building production systems end-to-end — from REST APIs and database architectures to real-time computer vision pipelines and fine-tuned LLMs. Comfortable working across the stack in TypeScript, Python, and Java, with hands-on deployment experience using Docker and AWS. AWS Certified in Machine Learning Engineer Associate, Solutions Architect Associate, and AI Practitioner.

Experience

ML Engineer Intern — February 2026 – Present
DevScope — Porto, Portugal (Hybrid)

- Developing a real-time computer vision system for automated people counting using YOLOv8-based object detection.
- Building and training deep learning models in PyTorch to detect and track individuals across video streams.
- Persisting tracking data to MongoDB and surfacing occupancy metrics through a Power BI dashboard for real-time and historical visibility.

Technologies: Python, PyTorch, YOLOv8, MongoDB, Power BI, Computer Vision

Education

BSc in Informatics Engineering
ISEP, Instituto Superior de Engenharia do Porto — Porto, Portugal

Technical Skills

Languages	Python, TypeScript, JavaScript, Java, C, C#, SQL, HTML/CSS
Frameworks	Node.js, Next.js, FastAPI, Spring Boot, Angular, .NET
ML / DL	PyTorch, Scikit-Learn, Pandas, Hugging Face Transformers, PEFT, QLoRA, MLflow, YOLOv8, DeepSORT
Cloud & DevOps	AWS, Docker, Git, Linux, CI/CD
Databases	PostgreSQL, MongoDB, Firebase, Oracle

Certifications

- AWS Certified Solutions Architect – Associate — *August 2025*
- AWS Certified Machine Learning Engineer – Associate — *June 2025*
- AWS Certified AI Practitioner – Foundational — *April 2025*
- IELTS Academic – C1 — *February 2026*

Projects

her. — *AI Personal Stylist — Full-Stack Web Application*

- Built end-to-end recommendation pipeline with user style profiling, contextual preferences, and feedback analysis backed by PostgreSQL with 8+ tables.

- Designed REST API and automated ETL workflows aggregating user feedback data daily; integrated Hugging Face ML models for style classification.
- Architected the full system from database schema to frontend, handling auth, data pipelines, and model integration in a single production deployment.

Technologies: TypeScript, Node.js, Next.js, PostgreSQL, Supabase, Hugging Face, REST API

curate. — *Fine-tuned Outfit Recommendation Model*

- Fine-tuned Mistral-7B using QLoRA on the Polyvore dataset (21k outfit examples) for garment compatibility and outfit recommendation.
- Built full data cleaning pipeline enforcing outfit validity rules: category filtering, structural constraints, and deduplication.
- Tracked experiments with MLflow across LoRA rank and learning rate configurations; served model via FastAPI endpoint containerised with Docker on AWS ECS.

Technologies: Python, PyTorch, Hugging Face PEFT, Mistral-7B, QLoRA, MLflow, FastAPI, Docker, AWS ECS

Transformer from Scratch — *Decoder-only Music Generation Model*

- Built a decoder-only transformer from first principles in PyTorch — multi-head self-attention, positional encoding, residual connections, and layer norm — trained on MIDI data for symbolic music generation.
- Designed a custom tokenisation vocabulary of 388 tokens; built the full data pipeline from raw MIDI files using `pretty_midi`.
- Trained with cosine LR annealing, gradient clipping, and early stopping on perplexity; generation samples autoregressively with multinomial sampling.

Technologies: Python, PyTorch, Transformers, NLP, pretty_midi, MIDI

headsup — *Real-Time People Counting System* *(developed during internship at DevScope)*

- Designed a real-time computer vision system for automated people counting using YOLOv8-based object detection.
- Persisted tracking data to MongoDB and surfaced occupancy metrics through a Power BI dashboard for real-time and historical visibility.

Technologies: Python, PyTorch, YOLOv8, MongoDB, Power BI, Computer Vision

LSTM Sentiment Analyser — *Review Analyser*

- Trained a 2-layer LSTM on 25,000 IMDB reviews for binary sentiment classification achieving 80% accuracy; built custom tokenisation pipeline, embedding layer, and FastAPI serving layer deployed on Hugging Face Spaces.

Technologies: Python, PyTorch, LSTM, FastAPI, Hugging Face Spaces